



5486 - How do galaxies move their metals? Unraveling $z > 2$ disks with controlled numerical experiments tailored to JWST

Cycle: 3, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

Metals are Nature's tracer fluids, mapping gas flows in to, out of, and within galaxies as they grow over time. Imaging spectroscopy, such as JWST's unique NIR IFUs, can measure resolved metallicities inside galaxies during the first formation of disks at $z > 2$. Surprisingly, a large fraction of high- z metallicity gradients are positive, indicating that galaxy disks are more metal-rich in their outskirts relative to the center, in strong tension with theoretical expectations. We propose new cosmological zoom simulations optimized for understanding the rapid evolution of metallicity gradients in $z > 2$ disks. Our new simulations will make three advances over existing work. We will: (1) use high spatial resolution (< 100 pc) and time resolution (2 Myr) to track metals as they migrate within the disk into the CGM and back, (2) implement experimental controls over the relevant physics to isolate the key physical influences on metallicity gradients, and (3) release public data products to support current and future JWST datasets, including novel non-parametric metallicity maps and metallicity-velocity maps that can be compared to JWST's unique IFU spectroscopy.

OBSERVING DESCRIPTION

This is a theory proposal. No observations necessary.